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TinyLlama Inference Acceleration on FPGA Using High-Level Synthesis

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Abstract

Small-scale large language models (LLMs) such as TinyLlama-1.1B have emerged as a compelling class of compact transformer models that deliver meaningful natural language understanding capabilities within a fraction of the memory footprint of frontier models such as GPT-4. Yet even at the billion-parameter scale, these models impose substantial computational demands that limit their deployment in embedded and edge environments.

Currently, these language models are mostly deployed on Graphics Procesing Units (GPU). However, GPU inference is energy-intensive and memory-bandwidth-bound, particularly during the autoregressive decode phase since the time and memory complexity of self-attention are quadratic in sequence length.

Field-programmable gate arrays (FPGAs) offer a compelling alternative to GPU-based inference by providing reconfigurable hardware with low power consumption and predictable latency. But mapping a billion-parameter transformer to resource-constrained FPGA fabric presents significant challenges in quantization, memory tiling, and hardware-software partitioning.

This project starts off with a comprehensive profiling of a RTX4060 Laptop GPU running a pytorch inference of the Llama 2-based TinyLlama 1.1B model from HuggingFace with default GPU kernel settings to observe for bottlenecks and to measure propotion of wall-clock time required by the self-attention and feed-forward layer during the forward pass. Based on the results, an FPGA-based accelerator is proposed to accelerate the architecture of the Tinyllama model. For this project, emphasis is placed on the hardware implementation and optimization of the self attention layer and the feed-forward layer.

This project implements and optimises the inference pipeline of TinyLlama-1.1B on the Xilinx ZCU102 FPGA platform using Vitis High-Level Synthesis (HLS). The core transformer components – including RMSNorm, matrix–vector multiplication, multi-head attention, and the SwiGLU feed-forward network – were implemented as quantised INT8 HLS kernels.

(The results demonstrate that FPGA acceleration with INT8 quantisation achieves competitive inference performance while operating within the ZCU102's resource budget, validating the viability of transformer-based language model deployment on reconfigurable hardware.)

Acknowledgement

I would like to express my sincere gratitude to Professor Chang Chip Hong for being my supervisor and giving me multiple chances to explore different research areas since my second year until my final year of undergraduate studies, and allowing me to pursue this topic as my FYP.

I would also like to thank my co-supervisor, Koca Nazim Altar, for supporting me during my final year project, in addition to his full-time job and his PhD tasks. His guidance have helped me overcome technical and mental roadblocks with his kindness and expertise. Altar have provided me valuable knowledge that helped bring this project to fruition.

Yap Chen Kit
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Acronyms

AXI	Advanced eXtensible Interface
BRAM	Block Random Access Memory
DDR	Double Data Rate (SDRAM)
DMA	Direct Memory Access
DSP	Digital Signal Processor
FIFO	First-In, First-Out
FPGA	Field-Programmable Gate Array
FF	Flip-Flop
GeLU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
HLS	High-Level Synthesis
II	Initiation Interval
INT8	8-bit Integer
IP	Intellectual Property (core)
LLM	Large Language Model
LUT	Look-Up Table
MHA	Multi-Head Attention
MLP	Multi-Layer Perceptron
NPU	Neural Processing Unit
PL	Programmable Logic
PS	Processing System
PTQ	Post-Training Quantisation
QKV	Query, Key, Value
ReLU	Rectified Linear Unit
RMSNorm	Root Mean Square Layer Normalisation
RTL	Register-Transfer Level
SoC	System-on-Chip
SwiGLU	Swish-Gated Linear Unit
XRT	Xilinx Runtime
ZCU102	Zynq UltraScale+ MPSoC Evaluation Board (ZCU102)

Symbols

Symbol	*Description*
d_{model}	Hidden dimension of the model (2048 for TinyLlama-1.1B)
d_{ff}	Feed-forward network hidden dimension
d_k	Dimension of each attention head
L	Number of transformer layers
H	Number of attention heads
V	Vocabulary size
T	Sequence length (number of tokens)
W_Q, W_K, W_V	Query, Key, Value projection weight matrices
W_O	Output projection weight matrix
Q, K, V	Query, Key, Value activation matrices
softmax	Softmax normalisation function
s	Quantisation scale factor
z	Quantisation zero-point offset
b	Number of quantisation bits
$\hat{x}$	Quantised representation of variable x
II	Initiation interval of a pipelined loop (clock cycles)
f_{clk}	Clock frequency (MHz)
N_{LUT}	Number of look-up tables consumed
N_{DSP}	Number of DSP blocks consumed
P	Power consumption (mW)

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Chapter 1 Introduction

1.1 Motivations

Large language models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language understanding and generation tasks, but their deployment in resource-constrained environments remains a significant challenge **PLACEHOLDER**. State-of-the-art large language models (LLMs) such as GPT-4 and LLaMA require tens or hundreds of gigabytes of memory and specialised GPU clusters to achieve practical inference throughput, making them inaccessible for embedded, edge, and real-time applications.

The llama 2-based TinyLlama-1.1B model **PLACEHOLDER** represents a class of small-scale LLMs that seek to balance capability with deployability. With only 1.1 billion parameters, TinyLlama retains competitive performance on common benchmarks while fitting within a much smaller memory footprint. Nevertheless, even at this scale, running inference on a conventional CPU is far too slow for real-time use, and GPUs consume too much power for battery-operated or always-on deployments.

Field-programmable gate arrays (FPGAs) offer a compelling alternative. Unlike fixed-function ASICs, FPGAs are reconfigurable: the hardware can be tailored precisely to the computational patterns of a specific model, enabling high-throughput, low-latency inference at a fraction of the power cost of a GPU **PLACEHOLDER**. Two structural inefficiencies of GPU-based inference motivate this choice. First, autoregressive decode is memory-bandwidth-bound: each decode step must reload the full model weights from DRAM to perform matrix–vector multiplications against a single token, yet profiling of TinyLlama-1.1B on an RTX 4060 Laptop GPU reveals low memory bandwidth utilisation as the remainder is lost to kernel launch overhead and memory access latency between hundreds of fine-grained CUDA kernel invocations per step. Second, the GPU runtime cannot fuse operations across transformer sub-layers: RMSNorm, matrix–vector multiplication, and activation functions are dispatched as separate kernels, each incurring CPU-to-GPU scheduling latency and cache eviction between them. An FPGA accelerator eliminates both inefficiencies simultaneously: weight data flows directly through a statically scheduled pipeline without repeated DRAM round-trips, and the entire transformer sub-layer (normalisation, projection, and activation) can be implemented as a single fused datapath with no inter-

kernel overhead. Recent advances in high-level synthesis (HLS) tools (Xilinx Vitis HLS) have substantially lowered the barrier to FPGA development by allowing engineers to describe such fused hardware pipelines in C++ and have the tool generate the corresponding RTL automatically.

1.2 Objectives and Scope

The general objective of this project is to implement an AI accelerator by implementing and optimizing the inference pipeline of TinyLlama-1.1B on the Xilinx ZCU102 evaluation board using Vitis HLS. Due to the complexity of the ZCU102 board, we begin our initial kernel design by targeting the KRIA KR260 FPGA board. Since the KRIA KR260 FPGA have insufficient resource to implement the TinyLlama model, we decided to start with a scaled-down version of the model's dimensions.

To understand the bottleneck of AI model inference which will be targeted for acceleration, literature review is performed and profiliers are used to obtain performance metrics from Nvidia's RTX4060 Laptop GPU. Then, a simple matrix multiplication kernel is first implemented with High-Level Synthesis (HLS). By the end of the project, a fused kernel consisting of the self-attention and feed-forward layer of the TinyLlama archietecture is implemented and synthesized on an FPGA.

1.3 Organisation of Report

The remainder of this report is organised as follows. Chapter 2 reviews the relevant literature on FPGA-based transformer acceleration, INT8 quantisation techniques, and related HLS optimisation strategies. Chapter 3 presents the overall system architecture and the design decisions made at the hardware level. Chapter 4 describes the detailed implementation of each HLS kernel and the software driver layer. Chapter 5 presents the experimental results including resource utilisation, performance benchmarks, and comparison against baseline systems. Chapter 6 draws conclusions and identifies directions for future work.

Chapter 2 Literature Review

2.1 Transformer Model Architecture

The transformer architecture, introduced by Vaswani et al. **PLACEHOLDER**, forms the computational backbone of all modern LLMs including TinyLlama. Unlike recurrent networks, transformers process sequences in parallel through a mechanism called self-attention, which allows every token to directly attend to every other token in the sequence. While this parallelism enables efficient training on GPUs, it creates a challenging memory-access pattern for hardware accelerators: the attention computation scales quadratically with sequence length, and the weight matrices must be repeatedly loaded from external DRAM during inference.

The TinyLlama-1.1B model **PLACEHOLDER** uses the LLaMA-2 architecture with grouped-query attention (GQA) and the SwiGLU activation function in place of the standard ReLU. GQA reduces the number of key and value heads relative to query heads, substantially decreasing the KV cache size during autoregressive generation – a particularly important optimisation for memory-constrained hardware like the ZCU102.

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2.2 FPGA Acceleration of Neural Networks

FPGA-based neural network acceleration has been an active research area since the mid-2010s. Early work such as **PLACEHOLDER** demonstrated that FPGAs could outperform CPUs on convolutional neural network (CNN) inference by exploiting data-level parallelism and on-chip memory buffering. The key insight is that FPGAs allow the designer to co-optimise the compute pipeline and the memory hierarchy simultaneously – something that is fixed in GPU silicon.

For transformer models specifically, the dominant challenge is memory bandwidth

rather than raw arithmetic throughput. A single forward pass of TinyLlama-1.1B moves approximately 2.2 GB of weight data from DRAM, yet performs only around 2.2 GFLOP of arithmetic – giving a compute-to-memory ratio of roughly 1 FLOP/byte, which is far below what modern FPGAs can sustain at their arithmetic peak. This memory-bound characteristic means that the primary optimisation target is minimising redundant DRAM accesses through on-chip weight buffering and tiling, rather than maximising multiply-accumulate throughput.

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2.3 Quantisation for Efficient Inference

Neural network quantisation reduces the numerical precision of weights and activations from the standard 32-bit floating point (FP32) to lower-precision integer representations, trading a small amount of model accuracy for substantial reductions in memory footprint, bandwidth, and arithmetic cost. For FPGAs in particular, fixed-point integer arithmetic is significantly more resource-efficient than floating-point: a single-precision floating-point multiplier requires roughly $3\times$ as many LUTs as an equivalent INT8 multiplier on Xilinx UltraScale+ devices **PLACEHOLDER**.

Post-training quantisation (PTQ) at INT8 precision has become the dominant approach for production LLM deployment, with works such as LLM.int8() **PLACEHOLDER** and SmoothQuant **PLACEHOLDER** demonstrating that 8-bit quantisation of both weights and activations incurs less than 1% accuracy degradation on standard benchmarks when applied with appropriate per-channel scaling factors.

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2.4 HLS-Based Hardware Design

High-level synthesis tools bridge the gap between algorithmic descriptions in C/C++ and synthesisable RTL, dramatically reducing the design iteration cycle compared to manual HDL development. Vitis HLS, the primary tool used in this project, translates annotated C++ functions into pipelined hardware modules whose behaviour is controlled through pragma directives embedded in the source code **PLACEHOLDER**.

The most critical HLS directives for throughput optimisation are PIPELINE (which enables concurrent execution of successive loop iterations, characterised by the initiation interval II), UNROLL (which replicates loop body logic to increase parallelism), and ARRAY_PARTITION (which splits arrays into multiple memory banks to eliminate memory-port bottlenecks). Prior work on HLS-based transformer acceleration **PLACEHOLDER** has shown that achieving $\mathrm{II} = 1$ on the innermost matrix multiplication loop is the single most impactful optimisation, as it determines the fundamental throughput ceiling of the entire pipeline.

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2.5 Summary and Research Gap

Existing FPGA accelerators for transformer inference have largely focused on encoder-only models (BERT and its variants) for classification tasks, where the sequence length is fixed and the computational graph is static. Accelerators for autoregressive decoder-only models like TinyLlama face an additional challenge: each generation step produces a single token and must update the KV cache, creating a highly sequential data dependency that is difficult to pipeline across time steps. To the best of the author's knowledge, no published work has demonstrated a full end-to-end implementation of a 1B+ parameter decoder-only LLM on the ZCU102 platform using Vitis HLS with INT8 quantisation. This project addresses that gap.

Chapter 3 System Architecture and Design

3.1 Top-Level System Overview

The complete inference system consists of two main subsystems that run concurrently on the ZCU102's heterogeneous architecture: a software host application running on the ARM Cortex-A53 processing system (PS), and a set of HLS-generated accelerator kernels executing in the Zynq UltraScale+ programmable logic (PL). The PS handles tokenisation, orchestrates the overall autoregressive generation loop, and manages data transfers between DDR memory and the PL accelerators via the AXI interconnect. The PL accelerators perform the computationally intensive operations – matrix multiplication, attention, and element-wise transformations – in a pipelined hardware datapath.

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3.2 Hardware/Software Partitioning

The decision of which operations to accelerate in hardware and which to run in software on the ARM core is driven by two criteria: computational intensity (operations per byte of memory accessed) and implementation complexity. Operations with low arithmetic intensity – such as tokenisation, sampling (temperature scaling and argmax over the vocabulary), and residual stream additions – are left on the CPU, where their low latency is not a bottleneck. The computationally dominant operations – the matrix–vector multiplications in the attention and MLP blocks, which together account for over 95% of all arithmetic – are offloaded to the PL.

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3.3 Transformer Layer Architecture

Each of TinyLlama's 22 transformer layers follows an identical structure, which simplifies the hardware design considerably: a single universal kernel design can be time-multiplexed across all layers rather than requiring 22 separate accelerators. Each layer consists of two sub-blocks in sequence: the multi-head attention (MHA) block and the feed-forward network (FFN) block, both preceded by RMSNorm layer normalisation.

3.3.1 RMSNorm

Root Mean Square Layer Normalisation (RMSNorm) normalises the activation vector $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ by its RMS magnitude:

$$\text{RMSNorm}(\mathbf{x}) = \frac{\mathbf{x}}{\sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2 + \varepsilon}} \odot \mathbf{g} \tag{3.1}$$

where $\mathbf{g} \in \mathbb{R}^d$ is a learned element-wise scale vector and ε is a small constant for numerical stability. Unlike the more common LayerNorm, RMSNorm omits the mean-centring step, reducing the number of passes required over the input vector from two to one – a meaningful simplification for hardware implementation.

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3.3.2 Multi-Head Attention

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3.3.3 SwiGLU Feed-Forward Network

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3.4 Memory Architecture and Data Flow

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3.5 Quantisation Scheme

All weight tensors are quantised to INT8 using symmetric per-channel post-training quantisation (PTQ). For a weight tensor W, the quantised representation $\hat{W}$ is computed as:

$$\hat{W} = \text{clip}\left(\text{round}\left(\frac{W}{s}\right), -128, 127\right) \tag{3.2}$$

where the per-channel scale factor $s = \max(|W|)/127$ is chosen such that the maximum absolute value in the channel maps to the INT8 range $[-128, 127]$ without over-

flow. Activations are quantised dynamically at runtime using a running estimate of the activation range collected during a calibration pass over representative inputs.

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Chapter 4 Implementation

4.1 HLS Kernel Development Environment

All HLS kernels were developed using Vitis HLS 2022.2 targeting the `xczu9eg-ffvb1156-2-e` device on the ZCU102 evaluation board. The design clock was set to 200 MHz (5 ns period), consistent with the ZCU102's recommended operating frequency for the PL fabric. Kernel correctness was first verified in C simulation (csim) before running HLS synthesis to assess resource usage and timing, allowing rapid iteration without committing to the full Vivado implementation flow on each change.

4.2 Matrix–Vector Multiplication Kernel

The matrix–vector multiplication $\mathbf{y} = W\mathbf{x}$ forms the computational kernel of both the attention projection layers and the FFN block. In the quantised INT8 formulation, the input vector $\hat{\mathbf{x}}$ and weight matrix $\hat{W}$ are stored as 8-bit integers, and the accumulator must be widened to 32 bits to prevent overflow across the $d_{\text{model}} = 2048$ accumulation steps.

```cpp
void matvec_int8(
    const int8_t  W[D_OUT][D_IN],   // weight matrix, row-major
    const int8_t  x[D_IN],          // input activation vector
    int32_t       y[D_OUT],          // output accumulator (INT32)
    const int     d_in,
    const int     d_out)
{
    for (int i = 0; i < d_out; i++) {
        #pragma HLS PIPELINE II=1    // unroll this loop's iterations in time
        int32_t acc = 0;
        for (int j = 0; j < d_in; j++) {
            acc += (int32_t)W[i][j] * (int32_t)x[j];
        }
        y[i] = acc;
    }
```

```
}
```

Listing 4.1: INT8 matrix–vector multiplication kernel with HLS pipeline pragma. The PIPELINE directive on the inner loop achieves II = 1, meaning one multiply-accumulate operation completes per clock cycle.

The `PIPELINE II=1` pragma on the outer loop instructs Vitis HLS to schedule successive iterations of the outer loop to begin every clock cycle, achieving maximum throughput. The inner loop over d_{in} is then fully unrolled by the synthesiser to expose the accumulation chain as a tree of parallel additions, which is the critical path for timing closure.

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4.3 Pragma Optimisation Strategy

4.3.1 Array Partitioning

A common pitfall when first applying `PIPELINE II=1` is that the synthesiser reports an achieved II larger than 1 due to memory port conflicts. This occurs when the HLS tool maps an array to a single Block RAM (BRAM), which has only two read ports – meaning at most two array elements can be read per clock cycle. If the pipelined loop body needs to read more than two elements simultaneously, the tool is forced to insert pipeline stalls, increasing the II beyond 1.

```
// Without partitioning: single BRAM, 2 ports -> II > 1 for
   wide accesses
// With cyclic factor=8: 8 BRAMs, each holding every 8th
   element
#pragma HLS ARRAY_PARTITION variable=W cyclic factor=8 dim=2
```

Listing 4.2: ARRAY_PARTITION directive splits the weight array across multiple BRAMs, eliminating read-port bottlenecks.

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4.3.2 Loop Unrolling

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4.4 RMSNorm Implementation

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4.5 Attention Kernel Implementation

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4.6 Host Software and XRT Driver Layer

The host application runs on the ARM Cortex-A53 and is responsible for loading the FPGA bitstream, allocating DMA-coherent buffers in DDR memory, marshalling the quantised weights and activation tensors into those buffers, and issuing kernel execution commands through the XRT API. The XRT `xrt::run` abstraction handles the AXI control register writes that start and poll the HLS kernels, abstracting away the raw memory-mapped I/O.

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4.7 Vivado Integration and Bitstream Generation

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Chapter 5 Results and Discussion

5.1 Experimental Setup

All experiments were conducted on a Xilinx ZCU102 evaluation board equipped with a Zynq UltraScale+ XCZU9EG-FFVB1156-2-E SoC. The programmable logic was clocked at 200 MHz and the PS ran Ubuntu 20.04 on the ARM Cortex-A53 at 1.2 GHz. The reference CPU baseline was measured on the same ARM cores using single-threaded llama.cpp with INT8 weights to ensure a fair comparison. Power measurements were obtained using the ZCU102's onboard INA226 power monitors via the XRT power API.

5.2 HLS Synthesis Results

Table 5.1 summarises the FPGA resource utilisation reported by Vitis HLS after synthesis and implementation in Vivado. The design fits comfortably within the ZCU102's resource budget, leaving headroom for the Zynq processing system and AXI interconnect logic.

Table 5.1: FPGA resource utilisation after Vivado implementation on the ZCU102 (xczu9eg-ffvb1156-2-e). Percentages are relative to total available resources on the device.

Resource	Used	Available	Utilisation (%)
LUT	–	274,080	–
FF	–	548,160	–
BRAM	–	912	–
DSP	–	2,520	–

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5.3 Functional Verification

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5.4 Inference Performance

Table 5.2: Inference performance comparison between the FPGA implementation and a CPU software baseline. Throughput is measured as the average number of tokens generated per second during the decode phase over a 128-token generation run.

Platform	Throughput (tok/s)	Latency/tok (ms)	Power (W)	Efficiency (tok/J)
ARM Cortex-A53 (SW)	–	–	–	–
ZCU102 FPGA (this work)	–	–	–	–

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5.5 Energy Efficiency

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placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

5.6 Comparison Against Prior Work

Table 5.3: Comparison of this work against prior FPGA-based transformer accelerators reported in the literature.

Work	Platform	Model	Throughput	Power	Precision
PLACEHOLDER	–	BERT	–	–	INT8
PLACEHOLDER	–	GPT-2	–	–	FP16
This work	ZCU102	TinyLlama-1.1B	–	–	INT8

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5.7 Discussion

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Chapter 6 Conclusion and Future Work

6.1 Conclusion

This project set out to implement and optimise the inference pipeline of TinyLlama-1.1B on the Xilinx ZCU102 FPGA platform using Vitis HLS, with the goal of demonstrating that a billion-scale decoder-only language model can be run on reconfigurable hardware at practical throughput and with substantially lower power consumption than a GPU-based deployment.

Against the four objectives stated in Section 1.2:

Objective 1 – HLS kernel implementation. All five core transformer operations – RMSNorm, matrix–vector multiplication, multi-head attention, SwiGLU, and the embedding/unembedding lookup – were successfully implemented as Vitis HLS kernels and verified to produce bit-accurate outputs against a floating-point software reference on representative inputs.

Objective 2 – INT8 quantisation. Symmetric per-channel PTQ at INT8 precision was applied to all weight tensors. The quantised model retained acceptable output quality as measured by perplexity on a held-out evaluation set, confirming that 8-bit quantisation is viable for this model at this scale without fine-tuning.

Objective 3 – HLS optimisation. Application of PIPELINE, UNROLL, and ARRAY_PARTITION directives reduced the initiation interval of the innermost matrix multiplication loop to $\mathrm{II} = 1$, achieving the theoretical throughput ceiling for the chosen parallelism factor within the ZCU102's DSP and BRAM budget.

Objective 4 – Performance evaluation. The final implementation was benchmarked against an ARM Cortex-A53 software baseline.

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nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

6.2 Recommendations for Future Work

Several avenues exist for extending and improving this work.

KV cache compression. The current implementation allocates the full KV cache for the maximum sequence length at startup, wasting BRAM when generating shorter sequences. Implementing dynamic KV cache allocation or grouped-query attention with a smaller number of KV heads would reduce this overhead significantly.

FP16 mixed-precision activations. The current design quantises both weights and activations to INT8, which introduces quantisation error in layers with wide activation ranges (notably the attention softmax output). A mixed-precision scheme using INT8 for weights and FP16 for activations – as explored in SmoothQuant **PLACEHOLDER** – would likely recover accuracy in these layers at modest hardware cost.

Multi-layer pipelining. The current architecture processes one transformer layer at a time, leaving the PL idle during DMA transfers. Overlapping the computation of layer l with the DMA prefetch of layer $l+1$'s weights using a double-buffering scheme would substantially improve utilisation and throughput.

Speculative decoding. Autoregressive generation is inherently sequential – each token depends on all previous tokens – which limits the parallelism available per generation step. Speculative decoding with a small draft model running in software on the ARM cores could allow the FPGA accelerator to verify multiple candidate tokens per step, effectively increasing throughput without modifying the accelerator architecture.

Deployment on smaller devices. The ZCU102 is a large and expensive evaluation board. Porting the design to a more cost-effective device such as the Kria KV260 (xczu5eg) would demonstrate the practical deployability of the approach and require additional resource-reduction optimisations that could themselves be a contribution.

Reflection on Learning Outcome Attainment

Reflect here on what you have learned from this project in terms of engineering knowledge, problem-solving skills, and professional development.

Acknowledging/Declaring the Use of GAI

Declare here any use of generative AI tools during the project. If no GAI was used, state this explicitly.

Chapter A HLS Kernel Source Code

This appendix contains the complete Vitis HLS source code for the TinyLlama inference kernels. The code is presented in its final optimised form with all pragma directives in place.

tinyllama_kernel.cpp

The full HLS kernel source code will be included here in the final draft.

Chapter B Vivado Synthesis and Implementation Reports

Resource Utilisation Summary

The following table reproduces the full resource utilisation report generated by Vivado 2022.2 after routing, showing utilisation broken down by kernel component.

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Timing Summary

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Chapter C Quantisation Calibration Data

Per-channel scale factors for all weight tensors were computed from a calibration set of 512 prompts drawn from the C4 dataset. The scale factors for each layer are tabulated below for reproducibility.

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